COMPLETION OF A (QUASI-)SEPARATED MORPHISM IS (QUASI-)SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-quasi-separated-gives-quasi-separated; the proof environment immediately following it is its own proof; the context span delivers the section situation that 'in the situation above' refers to

Editorial note. The label lemma-quasi-separated-gives-quasi-separated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the /T completion functor and its cartesian square (section-completion-functor setup, delivered as context)
- separatedness of a morphism of formal algebraic spaces via quasi-compactness resp. closed immersion of the diagonal (formal-spaces-section-separation-axioms)
- the remark identifying the diagonal of the completion morphism with the completion of the diagonal (remark-diagonal-gives-diagonal), and the two transfer lemmas of the same section: quasi-compact-gives-quasi-compact and closed-immersion-gives-closed-immersion

Source reference. Original lemma 0APV: sources/stacks-chow/tags.txt, source lines 13488–13488. Prerequisite pointer invoked at 6166; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0ARM: sources/stacks-chow/tags.txt, source lines 13550–13550. Prerequisite pointer invoked at 6114; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDC: sources/stacks-restricted/pinned-original-source.tex, source lines 6037–6037. Prerequisite pointer invoked at 6119; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDE: sources/stacks-restricted/pinned-original-source.tex, source lines 6067–6067. Prerequisite pointer invoked at 6118; it names

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a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDF: sources/stacks-restricted/pinned-original-source.tex, source lines 6091–6091. Prerequisite pointer invoked at 6117; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

1. The completion functor

In this section we consider the following situation. First we fix a base scheme S. All rings, topological rings, schemes, algebraic spaces, and formal algebraic spaces and morphisms between these will be over S. Next, we fix an algebraic space X and a closed subset $T \subset |X|$. We denote $U \subset X$ be the open subspace with $|U| = |X| \setminus T$. Picture

$$U \to X \quad |X| = |U| \amalg T$$

In this situation, given an algebraic space X' over X, i.e., an algebraic space X' endowed with a morphism $f : X' \to X$, then we denote $T' \subset |X'|$ the inverse image of T and we let $U' \subset X'$ be the open subspace with $|U'| = |X'| \setminus T'$. Picture

$$U' = f^{-1}U \qquad \begin{array}{ccc} U' & \longrightarrow & X' \\ \downarrow & & \downarrow{\scriptstyle f} \\ U & \longrightarrow & X \end{array} \qquad \begin{array}{ccccc} |U'| & \longrightarrow & |X'| & \longleftarrow & T' \\ \downarrow & & \downarrow{\scriptstyle |f|} & & \downarrow \\ |U| & \longrightarrow & |X| & \longleftarrow & T \end{array} \qquad T' = |f|^{-1}T$$

We will relate properties of f to properties of the induced morphism

$$f_{/T} : X'_{/T'} \longrightarrow X_{/T}$$

of formal completions. As indicated in the displayed formula, we will denote this morphism $f_{/T}$. We have already seen that $f_{/T}$ is representable by algebraic spaces in Formal Spaces, Lemma 0APV. In fact, as the proof of that lemma shows, the diagram

$$\begin{array}{ccc} X'_{/T'} & \longrightarrow & X' \\ {\scriptstyle f_{/T}}\downarrow & & \downarrow{\scriptstyle f} \\ X_{/T} & \longrightarrow & X \end{array}$$

is cartesian. Please keep this fact in mind whilst reading the lemmas stated and proved below.

Lemma 1.1. *In the situation above. If f is (quasi-)separated, then $f_{/T}$ is too.*

Proof. (Separation conditions on morphisms of formal algebraic spaces are discussed in Formal Spaces, Section 0ARM.) We have to show that if Δ_f is quasi-compact, resp. a closed immersion, then the same is true for $\Delta_{f_{/T}}$. This follows from the discussion in Remark 0GDF and Lemmas 0GDE and 0GDC. □